Inference

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Outline

Logical inference
- Set theory and logic
- Logical inference

Probability background
- Probability facts
- Conditional probability and independence
- Posterior distributions and model estimation
- Random variables, expectation and variance

Graphical models
- Graphical model
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Set theory

- First, consider some universal set Ω.
- A set A is a collection of points x in Ω.
- $\{x \in \Omega : f(x)\}$: the set of points in Ω with the property that $f(x)$ is true.

Unary operators

- $\neg A = \{x \in \Omega : x \notin A\}$.

Binary operators

- $A \cup B$ if $\{x \in \Omega : x \in A \vee x \in B\}$ - (c.f. $A \vee B$)
- $A \cap B$ if $\{x \in \Omega : x \in A \wedge x \in B\}$ - (c.f. $A \wedge B$)

Binary relations

- $A \subset B$ if $x \in A \Rightarrow x \in B$ - (c.f. $A \implies B$)
- $A = B$ if $x \in A \Leftrightarrow x \in B$ - (c.f. $A \Leftrightarrow B$)

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The inference problem

- Given statements $A_1, \ldots, A_n$ we know to be true (i.e. a knowledge base), is another statement B true?

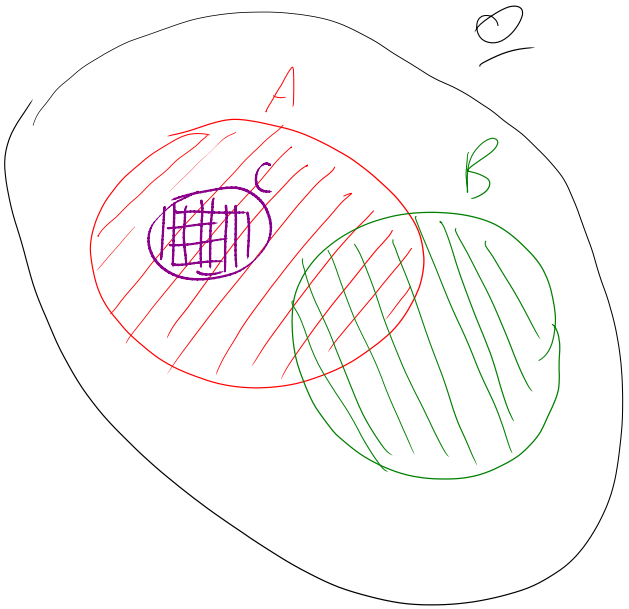
The following statements are equivalent:

- $A \implies B$ iff $(A \cap \neg B) = \varnothing$.
- $A \implies B$ iff $A \subset B$.

In addition

- If $(A \Rightarrow B) \wedge A$ then B.
- If $(A \wedge B)$ then A.

Illustration



$(A \mid C) =$

infered known

$(B \mid C) =$

$(C \mid A) =$

$(A \wedge B \mid C)$

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Events as sets

The universe and random outcomes

- The Ω contains all events that can happen.
- When something happens, we observe an element $\omega \in \Omega$.

Events in the universe

- An event is true if $\omega \in A$, and false if $\omega \notin A$.
- The negative event $\neg A = \Omega \setminus A$ is the set $\{\omega \in \Omega \wedge \omega \notin A$.
- The possible events are a collection of subsets Σ of Ω so that

(i) $\Omega \in \Sigma$, (ii) $A, B \in \Sigma \Rightarrow A \cup B \in \Sigma$ (iii) $A \in \Sigma \Rightarrow \neg A \in \Sigma$

Example: Traffic violation

- A car is moving with speed $\omega \in [0, \infty)$ in front of the speed camera.
- $A_0 = [0, 50]$: below the speed limit
- $A_1 = (50, 60]$: low fine
- $A_2 = (60, \infty]$: high fine
- $A_3 = (100, \infty)$: Suspension of license

All combinations of the above events are interesting.

Probability fundamentals

Probability measure P

Probability can be seen as an area-like function assigning a likelihood to sets.

- $P : \Sigma \to [0, 1]$ gives the likelihood $P(A)$ of an event $A \in \Sigma$.
- $P(\Omega) = 1$
- For $A, B \subset \Omega$, if $A \cap B = \varnothing$ then $P(A \cup B) = P(A) + P(B)$.

Marginalisation

Partition

If $A_1, \ldots, A_n$ are a partition of B then:

- $A_j \cap A_i = \varnothing$ for $i \neq j$
- $\bigcup_{i=1}^{n} A_i = B$.

Marginalisation

If $A_1, \ldots, A_n \subset \Omega$ are a partition of Ω

$$P(B) = \sum_{i=1}^{n} P(B \cap A_i).$$

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Conditional probability

Definition (Conditional probability)

The conditional probability of an event A given an event B is defined as

$$P(A|B) \triangleq \frac{P(A \cap B)}{P(B)}$$

The above definition requires $P(B)$ to exist and be positive.

Conditional probabilities as a collection of probabilities

More generally, we can define conditional probabilities as simply a collection of probability distributions:

$$\{P_\theta : \theta \in \Theta\},$$

where Θ is indexing possible values of θ.

- θ is sometimes called the model or parameter

Conditional probability game

Cards

You are each given a pair of cards. Each card is either black (♠, ♣) or red (♦, ♥).
In particular, each card is the probability that you get

R	R
R	B
B	R
B	B

The theorem of Bayes

Theorem (Bayes's theorem)

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

The theorem of Bayes

Theorem (Bayes's theorem)

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

The general case

If $A_1, \ldots, A_n$ are a partition of Ω, meaning that they are mutually exclusive events (i.e. $A_i \cap A_j = \varnothing$ for $i \neq j$) such that one of them must be true (i.e. $\bigcup_{i=1}^{n} A_i = \Omega$), then

$$P(B) = \sum_{i=1}^{n} P(B|A_i)P(A_i)$$

and

$$P(A_j|B) = \frac{P(B|A_j)}{\sum_{i=1}^{n} P(B|A_i)P(A_i)}$$

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Bayes's theorem

As a conditional measure

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)} = \frac{P(B \mid A)P(A)}{P(B \mid A)P(A) + P(B \mid \neg A)P(\neg A)}$$

Bayes's theorem

As a conditional measure

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As a causal explanation

$$\mathbb{P}(\text{cause} \mid \text{effect}) = \frac{\mathbb{P}(\text{effect} \mid \text{cause})\,\mathbb{P}(\text{cause})}{\mathbb{P}(\text{effect})}$$

Bayes's theorem

As a conditional measure

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)} = \frac{P(B \mid A)P(A)}{P(B \mid A)P(A) + P(B \mid \neg A)P(\neg A)}$$

As a causal explanation

$$\mathbb{P}(\text{cause} \mid \text{effect}) = \frac{\mathbb{P}(\text{effect} \mid \text{cause})\,\mathbb{P}(\text{cause})}{\mathbb{P}(\text{effect})}$$

As model inference

- Prior $\beta(\theta)$
- Model class $\{P_\theta(\beta) : \theta \in \Theta\}$
- Data x

$$\beta(\theta \mid x) = \frac{P_\theta(x)\beta(\theta)}{\mathbb{P}_\beta(x)} = \frac{P_\theta(x)\beta(x)}{\sum_{\theta' \in \Theta} P_{\theta'}(x)\beta(\theta')}$$

Example: COVID symptoms

Activity (with playing cards or dice)

- Pick two (x, y) from 1 to 10.
- If ($x = 1$ and $y < 9$), or (x is even and $y \geq 9$), you have symptoms.
- Do you have COVID?

Example: COVID symptoms

Activity (with playing cards or dice)

- Pick two (x, y) from 1 to 10.
- If ($x = 1$ and $y < 9$), or (x is even and $y \geq 9$), you have symptoms.
- Do you have COVID?

Information

- 20% of people have COVID
- 50% of people with COVID have symptoms.
- 10% of people with no COVID have symptoms.
- If you do have symptoms, what are the chances you have COVID?

Example: COVID symptoms

Activity (with playing cards or dice)

- Pick two (x, y) from 1 to 10.
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- Do you have COVID?

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Formalisation

- Prior $P(C) = 0.1$:
- Likelihood: $P(S|C) = 0.5$, $P(S|\neg C) = 0.1$
- Posterior:

$$P(C|S) = \frac{P(S|C)P(C)}{P(S|C)P(C) + P(S|\neg C)P(\neg C)}$$

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Random variables

A random variable $f : \Omega \to \mathbb{R}$ is a real-valued function, with $\omega \sim P$.

The distribution of f

The probability that f lies in some subset $A \subset \mathbb{R}$ is

$$P_f(A) \triangleq P(\{\omega \in \Omega : f(\omega) \in A\}),$$

and we write $f \sim P_f$.

Shorthands for RV

- For RVs $f : \Omega \to \mathbb{R}$, we write $P(f \in A)$ to mean $P_f(A)$.
- For RVs $f : \Omega \to X$, where X is a finite set e.g. $\{1, 2, \ldots, n\}$, we write $P(f = x) = P_f(\{x\})$ for any $x \in X$.

Independence of random variables

Two RVs f, g are independent in the same way that events are independent.

$$P(f \in A \wedge g \in B) = P(f \in A)P(g \in B) = P_f(A)P_g(B).$$

In that sense, $f \sim P_f$ and $g \sim P_g$.

Formal definition

More specifically, we are measuring the set of ω values for which $f(\omega) \in A$ and $g(\omega) \in B$:

$$P(\{\omega : f(\omega) \in A, g(\omega) \in B\}) = P_f(A)P_g(B).$$

Shorthand notation

Since the above is very cumbersome, we usually just write that

$$P(f, g) = P(f)P(g)$$

for any two independent random variables f, g.

Expectation

Discrete probability space Ω

For any real-valued random variable $f : \Omega \to \mathbb{R}$, the expectation with respect to a probability measure P is

$$\mathbb{E}_P(f) = \sum_{\omega \in \Omega} f(\omega) P(\omega).$$

Continuous probability space

When Ω is continuous, we define the probability $P(A)$ of any subset $A \subset \Omega$ through a *probability density p:

$$P(A) = \int_A p(\omega) d\omega.$$

We can then define the expectation through the density:

$$\mathbb{E}_P(f) = \int_\Omega f(\omega) p(\omega) d\omega.$$

Properties of expectations

The law of the unconscious statistician

If $x : \Omega \to X$, with $X \subset \mathbb{R}$, and $\omega \sim P$, we can say that $x \sim P_x$, with $P_x(k) = P(\{\omega : x(\omega) = k\})$.

$$E_P(x) = \sum_{\omega \in \Omega} x(\omega) P_x(x) = \sum_{k \in X} k P_x(k).$$

Linearity of expectations

For any RVs x, y:

$$\mathbb{E}_P(x + y) = \mathbb{E}_P(x) + \mathbb{E}_P(y)$$

Properties of expectations

The law of the unconscious statistician

If $x : \Omega \to X$, with $X \subset \mathbb{R}$, and $\omega \sim P$, we can say that $x \sim P_x$, with $P_x(k) = P(\{\omega : x(\omega) = k\})$.

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Linearity of expectations

For any RVs x, y:

$$\mathbb{E}_P(x + y) = \mathbb{E}_P(x) + \mathbb{E}_P(y)$$

We give the proof for the discrete case:

$$\begin{aligned}\mathbb{E}_P(x + y) &= \sum_{\omega \in \Omega} [x(\omega) + y(\omega)] P(\omega) \\ &= \sum_{\omega \in \Omega} x(\omega) P(\omega) + \sum_{\omega \in \Omega} y(\omega) P(\omega) = \mathbb{E}_P(x) + \mathbb{E}_P(y)\end{aligned}$$

Distributions of multiple variables

The joint distribution $P(x, y)$

For two (or more) RVs $x : \Omega \to \mathbb{R}$, and $y : \Omega \to \mathbb{R}$, this is a shorthand for the distribution of $(x(\omega), y(\omega))$ when $\omega \sim P$. We can also use $P(x = i, y = j)$ for the probability that the two variables assume the values i, j respectively.

Independence

If x, y are independent RVs then $P(x, y) = P_x(x)P_y(y)$.

Correlation

If x, y are not correlated then $\mathbb{E}_P(xy) = \mathbb{E}(x)\,\mathbb{E}(y)$.

IID (Independent and Identically Distributed) random variables

A sequence x_t of r.v.s is IID if $x_t \sim P$ so that

$$(x_1, \ldots, x_t, \ldots, x_T) \sim P^T$$

i.e. a T-length sample is drawn from the product distribution $P^T = P \times P \times \cdots \times P$.

Conditional expectation

Conditional expectations are similar to conditional probabilities.

Discrete Ω

The conditional expectation of a random variable $f : \Omega \to \mathbb{R}$, with respect to a probability measure P conditioned on some event B is simply

$$\mathbb{E}_P(f|B) = \sum_{\omega \in \Omega} f(\omega) P(\omega|B).$$

Conitnuous Ω

The conditional expectation of a random variable $f : \Omega \to \mathbb{R}$, with respect to a probability density p conditioned on some event B is simply

$$\mathbb{E}_p(f|B) = \int_\Omega f(\omega) p(\omega|B) d\omega.$$

Joint and conditional probabilities of RVs

Similarly to the notation over sets,

$$P(A \cap B) = P(A \mid B)P(B),$$

when dealing with RVs, it is common to use the notation

$$P(x, y) = P(x|y)P(y)$$

This equation works for all possible values of x, y e.g.

$$P(x = 1, y = 0) = P(x = 1|y = 0)P(y = 0)$$

which then denotes the probability msas of each

Probability notation: math versus statistics

- $P(C)$: Probability of event C
- $P(A \cap B)$: Probability of A and B
- $P(A|B)$: Probability of the event A if we know B
- $P(A \cup B)$: Probability of A or B

- $P(x)$: distribution of variable x.
- $P(x, y)$: joint distribution of x, y
- $P(x|y)$: distribution of x for different values of y
- No correspondence.

Example: The k-meteorologists problem (set notation)

- R_t: The event that it rains at time t.
- A set of stations Θ, with $\theta \in \Theta$ making weather predictions:

$$P(R_{t+1} \mid R_1, \ldots, R_t, \theta),$$

- A prior probability $P(\theta)$ on the stations.
- The marginal probability

$$P(R_1 \cap \cdots \cap R_t) = \sum_{\theta \in \Theta} P(R_1 \cap \cdots \cap R_t \mid \theta) P(\theta)$$

- The posterior probability

$$P(\theta \mid R_1 \cap \cdots \cap R_t) = \frac{P(R_1 \cap \cdots \cap R_t \mid \theta) P(\theta)}{P(R_1 \cap \cdots \cap R_t)} = \frac{\prod_{i=1}^{t} P(R_t \mid R_1 \cap \cdots \cap R_{t-1}}{P(R_1 \cap \cdots \cap R_t)}$$
$$= \frac{P(R_t \mid R_1 \cap \cdots \cap R_{t-1} \mid \theta) P(\theta \mid R_1 \cap \cdots \cap R_{t-1})}{P(R_t \mid R_1 \cap \cdots \cap R_{t-1})}$$

- The marginal posterior probability

$$P(R_{t+1} \mid R_1 \cap \cdots \cap R_t) = \sum_{\theta \in \Theta} P(R_{t+1} \mid R_1 \cap \cdots \cap R_t, \theta) P(\theta \mid R_1 \cap \cdots \cap R_t)$$

Example: The k-meteorologists problem (stat notation)

- $x_t \in \{0, 1\}$: A random variable, telling us whether it rains at time t.
- A set of stations Θ, with $\theta \in \Theta$ making weather predictions:

$$P_\theta(x_{t+1} \mid x_1, \ldots, x_t)$$

- A prior probability $\beta(\theta)$ on the stations.
- The marginal probability

$$\mathbb{P}_\beta(x_1, \ldots, x_t) = \sum_{\theta \in \Theta} P_\theta(x_1, \ldots, x_t)\beta(\theta)$$

- The posterior probability

$$\beta(\theta \mid x_1, \ldots, x_t) = \frac{P_\theta(x_1, \ldots, x_t)\beta(\theta)}{\mathbb{P}_\beta(x_1, \ldots, x_t)} = \frac{\prod_{i=1}^{t} P_\theta(x_t \mid x_1, \ldots, x_{t-1})\beta(\theta)}{\mathbb{P}_\beta(x_1, \ldots, x_t)}$$
$$= \frac{P_\theta(x_t \mid x_1, \ldots, x_{t-1})\beta(\theta \mid x_1, \ldots, x_{t-1})}{\mathbb{P}_\beta(x_t \mid x_1, \ldots, x_{t-1})}$$

- The marginal posterior probability

$$\mathbb{P}_\beta(x_{t+1} \mid x_1, \ldots, x_t) = \sum_{\theta \in \Theta} P_\theta(x_{t+1} \mid x_1, \ldots, x_t)\beta(\theta \mid x_1, \ldots, x_t)$$

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Graphical models

- A graphical model or Bayesian network is used to model dependencies between random variables.
- Each node in the graph is a variable.
- The arcs show what variable is an input to which variable.

Independence

Independent events $A \perp\!\!\!\perp B$

- A, B are independent iff $P(A \cap B) = P(A)P(B)$.
- Knowing if A happened, does not tell us anything about whether B happened

Conditional independence $A \perp\!\!\!\perp B \mid C$

- A, B are conditionally independent given C iff $P(A \cap B|C) = P(A|C)P(B|C)$.
- Knowing if C happened tells us all we need to know about A and B.

For random variables

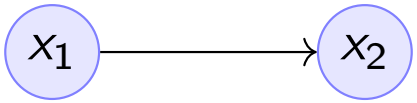
- Independence: $P(x, y) = P(x)P(y)$.
- Conditional independence: $P(x, y|z) = P(x|z)P(y|z)$.

Model specification: Independent

$$f = \text{Bernoulli}(1/2)$$
$$g = \text{Bernoulli}(0.8)$$
$$x_1 \sim f$$
$$x_2 \sim g$$

```
def f():
  return np.random.choice(2)
def g:
  return np.random.choice(2, [0.2, 0.8])
x1 = f()
x2 = g()
```

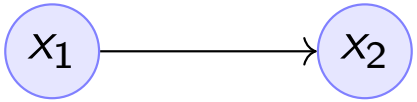
Model specification: Gaussian Dependent variables



$$f = \mathrm{Normal}(0, 1)$$
$$g(a) = \mathrm{Normal}(a, 1)$$
$$x_1 \sim f$$
$$x_2 | x_1 = a \sim g(a)$$

```
def f():
  return np.random.normal(0,1)
def g(a):
  return np.random.normal(a)
x1 = f()
x2 = g(x1)
```

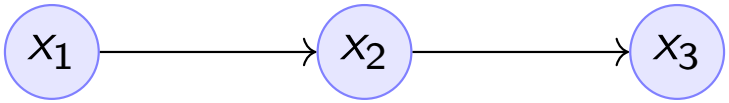
Model specification: Bernoulli Dependent variables



$$
\begin{aligned}
f &= \text{Bernoulli}(1/2) \\
g(a) &= \text{Bernoulli}(\theta_a) \\
x_1 &\sim f \\
x_2 | x_1 = a &\sim g(a) \\
\theta &= (0.6, 0.5)
\end{aligned}
$$

```
def f():
  return np.random.choice(2)
def g(a):
  theta = [0.6, 0.5]
  return np.random.choice(2,
 [1 - theta[a], theta[a]])
x1 = f()
x2 = g(x1)
```

Model specification: Chain



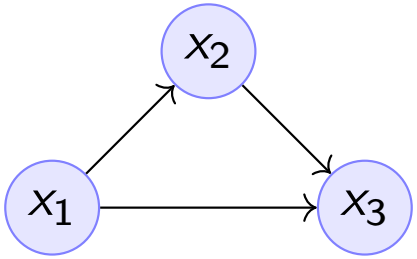
$$x_1 \sim f \tag{1}$$

$$x_2 \mid x_1 = a \sim g(a) \tag{2}$$

$$x_3 \mid x_2 = b \sim h(b), \tag{3}$$

```
def f():
  return np.random.uniform()
def g(a):
  return np.random.uniform() + a
def h(b)
  return np.random.uniform * b
x1 = f()
x2 = g(x1)
x3 = h(x2)
```

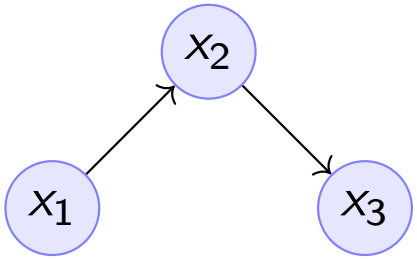
Graphical models



- Variables: x_1, x_2, x_3
- Arrows denote dependencies between variables.
- In this example the value of x_3 is a function of x_1, x_2, as well as a random input.

Conditional independence

Example



Graphical model for the factorisation $\mathbb{P}(x_3 \mid x_2)\,\mathbb{P}(x_2 \mid x_1)\,\mathbb{P}(x_1)$.

Definition

- Consider variables $x_1, \ldots, x_n$.
- Let B, D be subsets of $[n]$.

We say x_i is conditionally independent of $\boldsymbol{x}_B$ given $\boldsymbol{x}_D$ and write

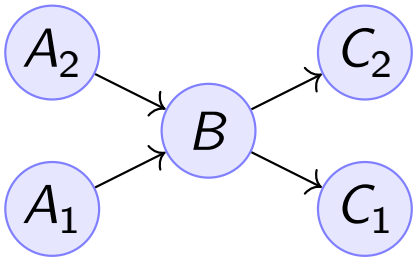
$$x_i \perp\!\!\!\perp \boldsymbol{x}_B \mid \boldsymbol{x}_D$$

if and only if:

$$\mathbb{P}(x_i, \boldsymbol{x}_B \mid \boldsymbol{x}_D) = \mathbb{P}(x_i \mid \boldsymbol{x}_D)\,\mathbb{P}(\boldsymbol{x}_B \mid \boldsymbol{x}_D).$$

Conditional independence

For any set of random variables $x_1, x_2, x_3, \ldots$, we can write their joint as $\prod_i P(x_i \mid x_1, \ldots, x_{i-1})$. However, we can use a Bayesian network to define conditional independence structures.



If A is a parent of B and C is a child of B, and there are no other paths from A to C then the following conditional independence holds:

$$P(C \mid B, A) = P(C \mid B)$$

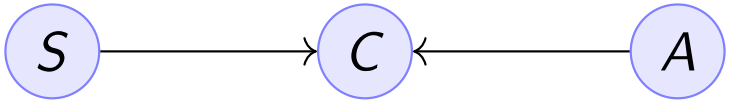
i.e. C is conditionally independent of A given B.

Conditional probability tables

We can now write the distribution of the above example as

$$P(B, C_1, C_2) = P(A_1)P(A_2)P(B|A_1, A_2)P(C_1|B)P(C_2|B).$$

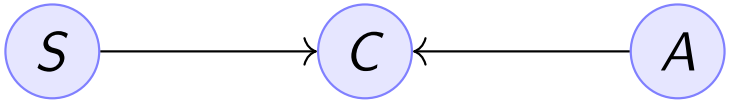
Smoking and lung cancer



Smoking and lung cancer graphical model, where S: Smoking, C: cancer, A: asbestos exposure.

- Here, S, A are independent
- However, they become dependent if we know C.

Smoking and lung cancer



Smoking and lung cancer graphical model, where S: Smoking, C: cancer, A: asbestos exposure.

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$$P(S, C, A) = P(S)P(A)P(C|S, A)$$

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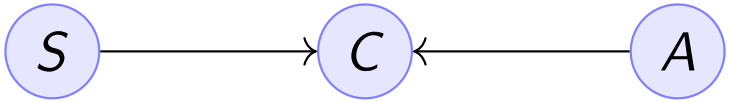

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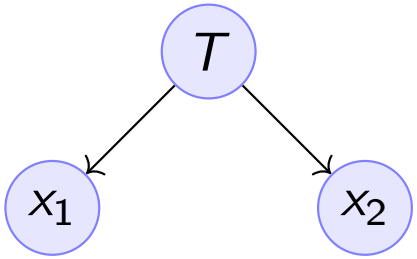
$$P(A, S|C) = P(A|S, C)P(S|C) = \frac{P(C|A, S)P(A|S)}{P(C|S)} \frac{P(C|S)P(S)}{P(C)}$$
$$= \frac{P(C|A, S)P(A|S)}{P(S|C)P(C)/P(S)} \frac{P(C|S)P(S)}{P(C)}$$

Time of arrival at work

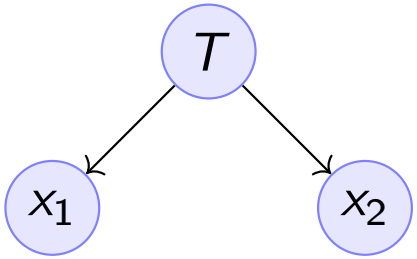


Time of arrival at work graphical model where T is a traffic jam and x_1 is the time John arrives at the office and x_2 is the time Jane arrives at the office.

*Conditional independence:

- Even though x_1, x_2 are not independent, they become independent once you know T.

Time of arrival at work



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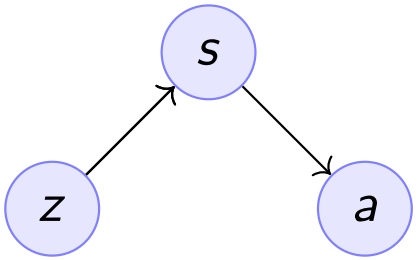
$$P(S, C, A) = P(S)P(A)P(C|S, A)$$

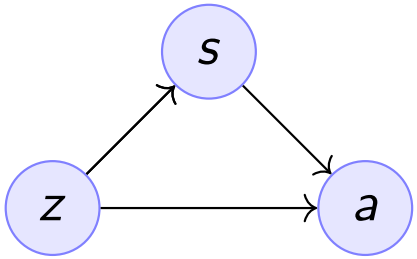
School admission

School	Male	Female
A	62	82
B	63	68
C	37	34
D	33	35
E	28	24
F	6	7

- z: gender
- s: school applied to
- a: admission



Is admission independent of gender?



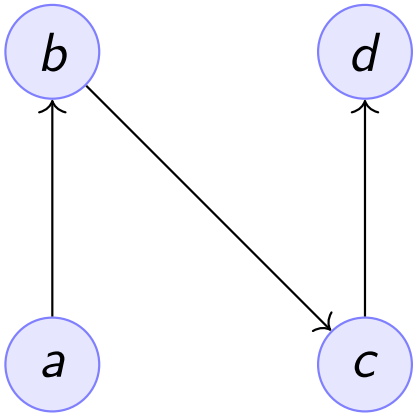
How about here?

Logical inference
Set theory and logic
Logical inference

Probability background
Probability facts
Conditional probability and independence
Posterior distributions and model estimation
Random variables, expectation and variance

Graphical models
Graphical model
Exercises

What is the model for this graph?



$$P(a, b, c, d) = \cdots$$

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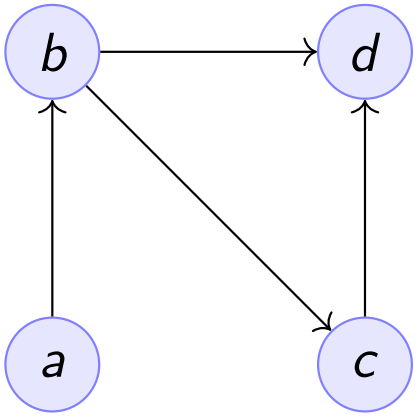

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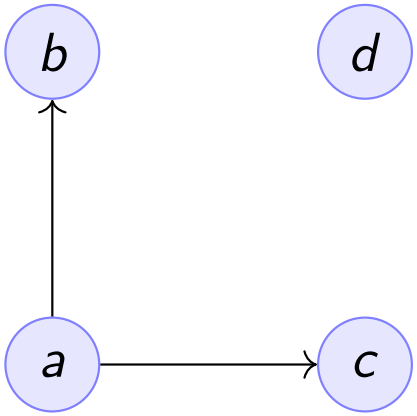
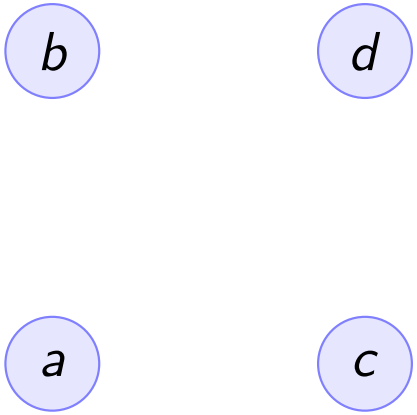


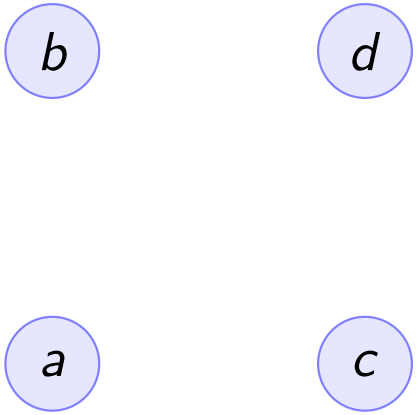
$$P(a, b, c, d) =$$

Draw the graph for this model



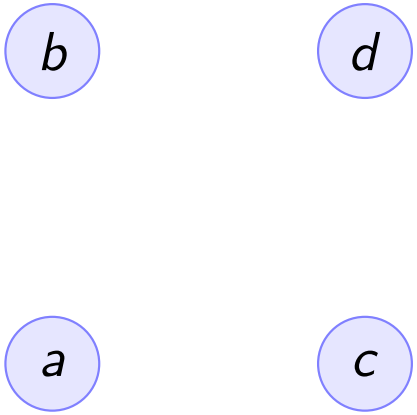
$$P(a, b, c, d) = P(a)P(b|a)P(c|b)P(d|b)$$

Draw the graph for this model



$$P(a, b, c, d) = P(a)P(b|a)P(d|c)P(c)$$

Draw the graph for this model



$$P(a, b, c, d) = P(a)P(b|a)P(c|a)P(d|b, c)$$

Example: COVID test

Information

- 10% of people have COVID
- 50% of people with COVID have a positive test
- 50% of people with COVID have symptoms
- 10% of people without COVID have a positive test
- 20% of people without COVID have symptoms

Example: COVID test

Information

- 10% of people have COVID
- 50% of people with COVID have a positive test
- 50% of people with COVID have symptoms
- 10% of people without COVID have a positive test
- 20% of people without COVID have symptoms

Formalisation

- Prior: $P(C = 1) = 0.1$
- Likelihood: $P(T, S|C) = P(T|C)P(S|C)$, $P(T, S|\neg C)$ for all va43lues of T, S, C.
- Posterior:

$$P(C|T, S) = \frac{P(S|C)P(T|C)P(C)}{\sum_{i=0}^{1} P(S|C = i)P(T|C = i)P(C = i)}$$

Example: Naive Bayes models

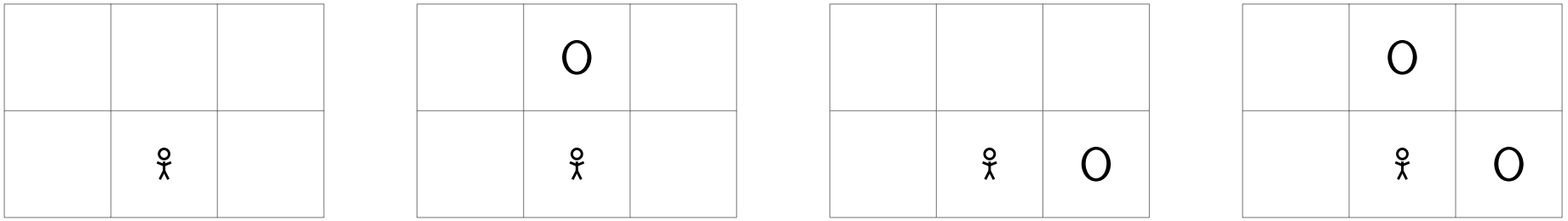
Sometimes we observe multiple effects that have a common cause, but which are otherwise independent:

$$\mathbb{P}(\text{effect}_1, \ldots \text{effect}_n \mid \text{cause}) = \prod_{i=1}^{n} \mathbb{P}(\text{effect}_i \mid \text{cause})$$

Naive Bayes model

- Observations $(\boldsymbol{x}_t, y_t)_{t=1}^{T}$ with $\boldsymbol{x}_t = (x_{t,1}, \ldots, x_{t,n})$.
- Probability models $P_\theta(y \mid \boldsymbol{x}) = \prod_{i=1}^{n} P_\theta(y \mid x_i)$.

Example: Wumpus world



Details

- Probability of each world A_i being true: $1/4$
- Probability of each hole generating a breeze: $P(B_1|A_2 \cup A_4) = P(B_2|A_3 \cup A_4)$ with B_1, B_2 conditionally independent given A.

Questions

- What is the probability of feeling a breeze $B = B_1 \cup B_2$ in each world?
- What is the probability of a hole above if you feel a breeze?
- What is the probability of a hole above f you don't feel a breeze?